

Part 5



The Plan!

□ Interventions

INTERVENTIONS

- Our account of causation involves causal assignments.
- A crucial feature of a causal assignment is that the values of variables are assigned, and not observed.
- In what follows, we examine what an *intervention* is and describe the kind that we consider ideal.

INTERVENTIONS

- **Ideal Interventions:**
 - An *ideal intervention* directly fixes the value of a variable within a causal system.
 - *Ideally intervening* to change the value of a cause does not change the relationship between the cause and its effects.
 - *Ideally intervening* to change the value of an effect does not change the relationship between the effect and its causes.

INTERVENTIONS

- **Ideal Interventions:**
 - Because ideal interventions can change the relationship between an effect and its causes, we distinguish between the causal system as it occurs *before* an intervention and the system as it occurs *after* an intervention.
 - **pre-intervention system** = “natural” or “pre-manipulated” system.
 - **post-intervention system** = “experimental,” “post-manipulation,” or “manipulated” system.

INTERVENTIONS

1. Intervention as Action
2. Ideal Interventions
3. Interventions and Causal Relationships
4. Pre-Manipulation vs. Post-Manipulation Graphs
5. Exogenous vs. Endogenous Variables
6. Interventions on Cyclic Causal Systems

INTERVENTIONS

1. Intervention as Action

- Sometimes a variable changes from naturally occurring causes, and sometimes we *act* to change it.
 - Intervention = Action

Example: Amount of money in your checking account, which can be categorized as high or low, depending on whether it is over or under 20,000 PKR



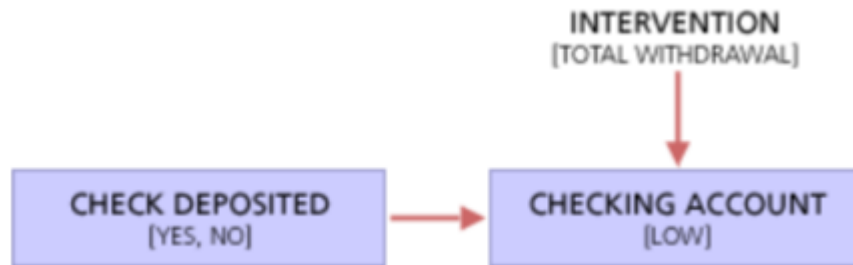
You can *intervene* from outside the system (Check Deposited Recently, Checking Account Level) to change the level of your checking account. You can:

- You can go to the bank and withdraw your entire account in cash (the total withdrawal intervention), or
- Deposit 20,000PKR in cash that you have stashed at home (the 20,000 PKR deposit intervention).

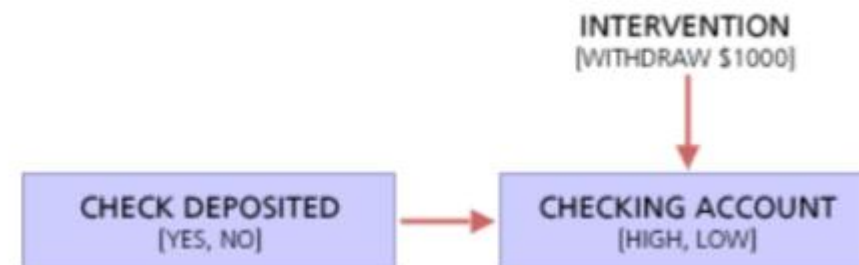
INTERVENTIONS

2. Ideal Interventions:

- is an action which
 1. directly targets exactly one variable in the causal system, and
 2. is completely effective in determining the value of the variable,
 3. and is itself exogenous, or uncaused.



Why could the example above violate the exogeneity assumption?



How does the second example differ from the first one?

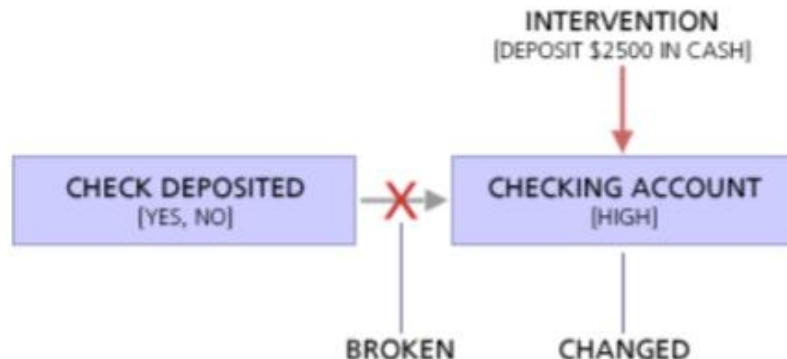
INTERVENTIONS

3. Interventions and Causal Relationships

- An ideal intervention on an effect does change the relationship between the effect and its causes.
- Example: The simple causal system involving your checking account:



- Whereas before we intervened on your paycheck, which is the cause, now we intervene on the effect (the checking account). Consider the 20,000 PKR deposit intervention.



The intervention changes the value of the Checking Account variable, but it also severs the relationship between the Paycheck variable and the Checking Account variable. The Paycheck deposit no longer has any influence on whether the account is high. The deposit 20,000 PKR in cash intervention took over all influence, and in so doing *changed the relationship between the effect and its causes*.

INTERVENTIONS

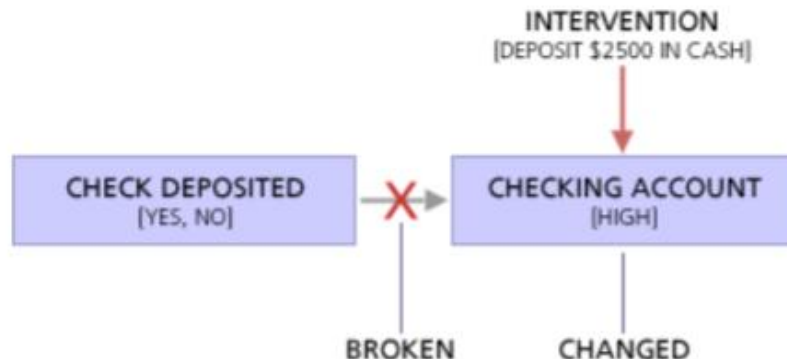
3. Interventions and Causal Relationships

- An ideal intervention is one that changes the value of the effect and its causes.
- Example: The sign on the door.

CHECK DEPOSITED
[YES, NO]

In general, if there is no cycle between a cause and its effect, then ideally intervening to change the value of the effect *breaks the relationship between the effect and its causes*.

- Whereas before we intervene on the effect (the checking account), consider the 20,000 PKR deposit intervention.



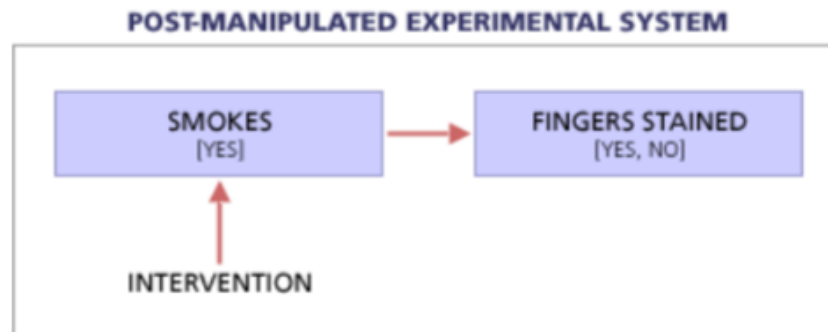
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INTERVENTIONS

- Pre-Manipulation vs. Post-Manipulation Graphs
 - As mentioned before, we call the pre-intervention system the “natural” or pre-manipulated” system, and the post-intervention system the “experimental” or “manipulated” system.



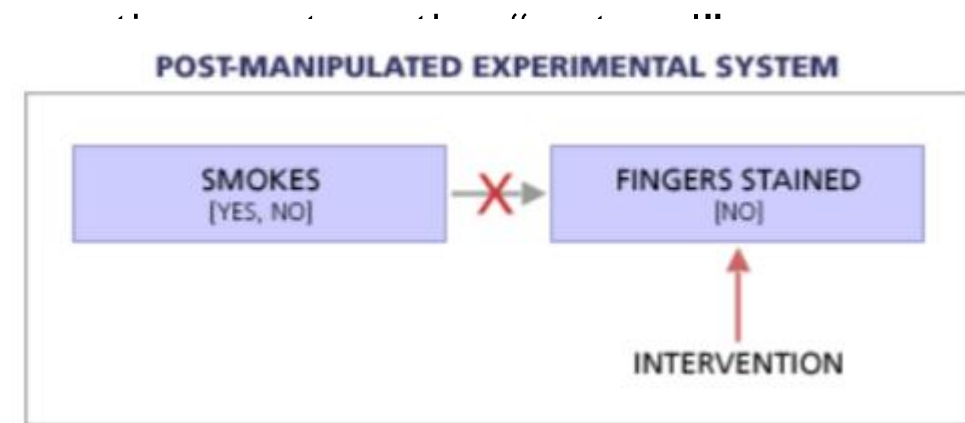
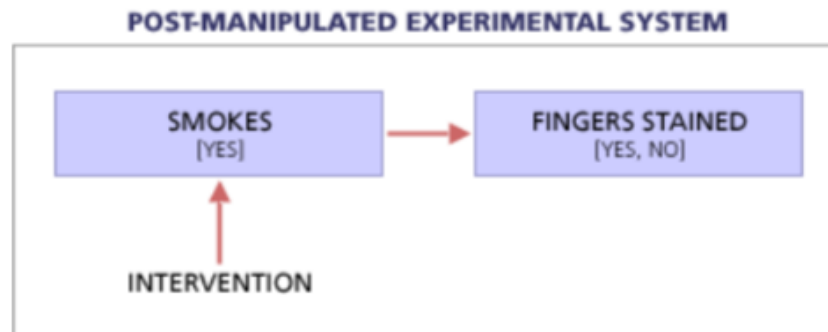
Smoking is a cause of tar stained fingers



In the second case, we intervene to prevent tar stains (e.g., we scrub our hands with tar solvent soap until they are stain free).

INTERVENTIONS

- Pre-Manipulation vs. Post-Manipulation Graphs
 - As mentioned before, we call the pre-interventor “pre-manipulated” system, and the post-interventor “manipulated” system.



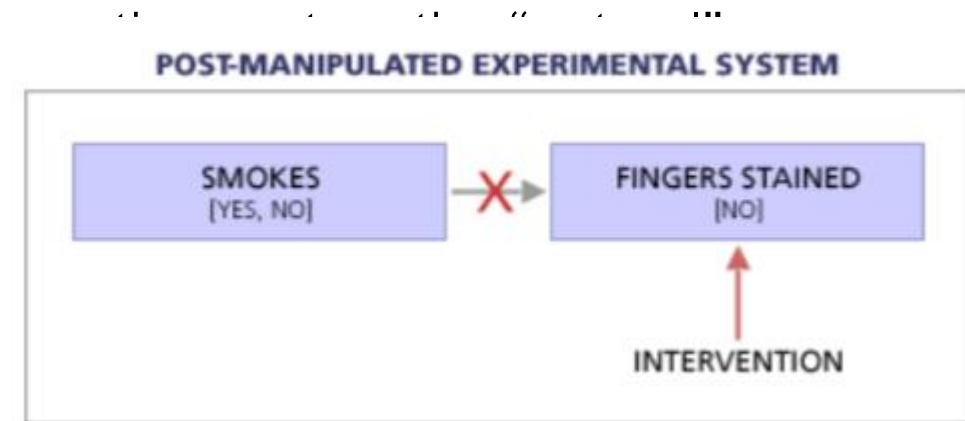
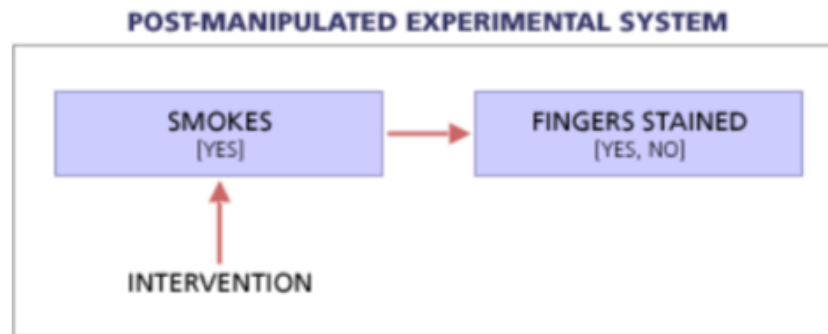
In the second case, we intervene to prevent tar stains (e.g., we scrub our hands with tar solvent soap until they are stain free).

- Influence of smoking on finger stains is broken.

INTERVENTION

To form the post-manipulated graph, begin with the pre-manipulated graph:

- Add the intervention as extra variable(s) to the causal graph with an arrow into the variable(s) it directly changes.
- Break or erase all the arrows pointing into variables that are directly intervened upon.
- Pre-Manipulation vs. Post-Manipulation
 - As mentioned before, we call the pre-intervened system, and the post-intervened “manipulated” system.



In the second case, we intervene to prevent tar stains (e.g., we scrub our hands with tar solvent soap until they are stain free).

- Influence of smoking on finger stains is broken.

INTERVENTIONS

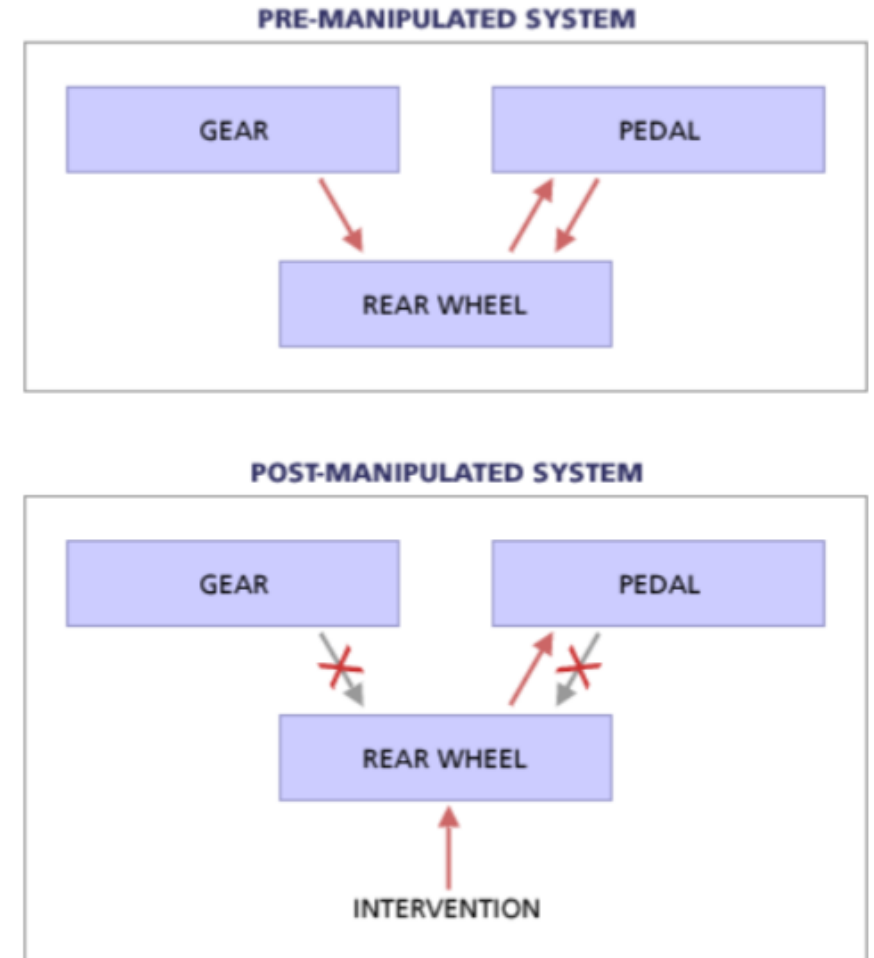
- 5. Exogenous vs. Endogenous Variables
 - It is useful to separate variables in causal systems into two sorts:
 - Independent, or exogenous variables, and
 - Dependent, or endogenous variables
- Independent, or Exogenous variables:
 - A variable is *independent, or exogenous*, if it is the effect of no other variable in the system.
- Dependent, or Endogenous variables:
 - A variable is *dependent, or endogenous*, if it is the effect of at least one other variable in the system.

INTERVENTIONS

- 5. Exogenous vs. Endogenous Variables
 - Because interventions only change the relationship between an effect and its causes, an intervention on an **exogenous** variable does not change the causal graph at all.
 - Except for adding the intervention variable, the pre-manipulation and post-manipulation causal graphs are the same.
 - We say that such interventions are *structure-preserving* (= if it is an intervention on **exogenous variables only**)
 - An (ideal) intervention on an **endogenous** variable does change the relationship between it and its causes.
 - We say that such an intervention is *structure-altering* (= if it is an intervention on **at least one endogenous variable**).

INTERVENTIONS

- 6. Interventions on Cyclic Causal Systems
 - The same rules for changing the causal graph from the pre-manipulated to the post-manipulated system apply to cyclic cases:
 - Add the intervention as an extra variable(s) to the causal graph.
 - Break or erase all the arrows pointing into variables that are directly intervened upon.



Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

A - Reflect and discuss – Interventions as Actions 1

 Question No 01 – Which of the following are actions under your direct control? (choose all that apply)

- A. Turning off the computer at which you are working.
- B. Removing corruption from politics.
- C. Not getting a cold.
- D. Taking 200% of the daily allowance of vitamin C.

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Answer - A and D.

Yes, turning off the computer at which you are working is an action under your direct control, and taking 200% of your daily allowance of vitamin C is an action under your direct control.

“Removing the corruption from politics” is too vague to even count as a hypothetical intervention, and “Not getting a cold” is not an action, it is an outcome, and it is not practically under your direct control.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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A - Reflect and discuss – Interventions as Actions 1

 **Question No 02 – Which of the following are practical or hypothetical interventions, as opposed to actions that are too vague to discuss sensibly? (choose all that apply)**

- A. Making all residents of New Jersey move to Nevada.
- B. Treating men and women the same.
- C. Eliminating racism.
- D. Putting a full time police officer in charge of guarding every apartment building in NYC.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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- C. Eliminating racism.
- D. Putting a full time police officer in charge of guarding every apartment building in NYC.

Answer - A and D.

Making all residents of New Jersey move to Nevada is not practical, but it is certainly clear enough to discuss hypothetically. Putting a police officer on every building in NYC is not practical, but it is certainly clear enough to discuss hypothetically. The other two choices are too vague.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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B - Reflect and discuss – Ideal Interventions 1

 **Question No 03** – Consider a system describing an airplane flying at cruising altitude with two variables:

1. Seat Belt Sign [lit, off]
2. Turbulence [yes, no]

If the pilot hears a weather report from a plane flying 100 miles ahead that turbulence is coming, and hits a switch to light up the Seat Belt Sign, this is not an ideal intervention on the Seat Belt Sign. **That is because: (choose one)**

- A. The intervention is not effective at determining the value of the variable it targets.
- B. The intervention does not directly target only the seat belt sign.
- C. The intervention is not exogenous to the system, it has been caused by some other variable, in this case a weather report of turbulence.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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B - Reflect and discuss – Ideal Interventions 1

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- A. The intervention is not effective at determining the value of the variable it targets.
- B. The intervention does not directly target only the seat belt sign.
- C. **The intervention is not exogenous to the system, it has been caused by some other variable, in this case a weather report of turbulence.**

Answer - C.

Not A: The intervention is effective at turning on the Seat Belt sign in a plane that is functioning normally.

Not B: The intervention will have no direct influence on whether the plane experiences turbulence, so it does only target a single variable.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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C - Reflect and discuss – Ideal Interventions 2

 **Question No 04 – Which of the following interventions would be completely effective at determining the value of the variable: Computer is turned on and functioning properly [yes, no]? (choose all that apply)**

- A. Sitting down for one hour and trying to make a computer function perfectly.
- B. Copying a file onto a computer that is infected with a computer virus.
- C. Removing the power source from a computer.
- D. Beating a computer with a sledgehammer for an hour.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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- B. Copying a file onto a computer that is infected with a computer virus.
- C. Removing the power source from a computer.**
- D. Beating a computer with a sledgehammer for an hour.**

Answer - C and D

Not A: Trying for an hour might help, but there is no guarantee that you could fix it if it was broken.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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C - Reflect and discuss – Ideal Interventions 2

Question No 05 – Suppose the causal system under consideration involves three variables: {Foot Speed, Endurance, Confidence}. **Which of the following interventions are directly targeted only at confidence, and are not “fat hand” interventions? (choose all that apply)**

- A. Hypnotizing the athlete into believing that he is great.
- B. Administering a drug that relieves anxiety but also relaxes muscles.
- C. Having the athlete watch videos of all his past performances that were successful.
- D. None of the above.

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- C. Having the athlete watch videos of all his past performances that were successful.**
- D. None of the above.

Answer - A and C.

Having the athlete watch videos or applying hypnosis should target only confidence directly.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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D - Reflect and discuss – Intervening on Causes 1

 **Question No 06** – Let's assume that high cholesterol causes poor artery condition. **If we ideally intervene to change your cholesterol, then... (choose one)**

- A. the relationship between Cholesterol Level and Artery Condition changes.
- B. ... the value of Cholesterol Level is determined by the intervention.
- C. ... the relationship between Cholesterol level and Heart Condition changes.
- D. None of the above.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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D - Reflect and discuss – Intervening on Causes 1

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- B. ... the value of Cholesterol Level is determined by the intervention.**
- C. ... the relationship between Cholesterol level and Heart Condition changes.
- D. None of the above.

Answer - B.

Not A: Since cholesterol level is a cause of artery condition, and we have intervened in cholesterol, we do not change the relationship between cholesterol level and artery condition.

Not C: Since cholesterol level is an indirect cause of heart condition, and we have intervened in cholesterol, we do not change the relationship between cholesterol level and heart condition.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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D - Reflect and discuss – Intervening on Causes 1

 Question No 07 – In general, an ideal intervention on a cause...

- A. determines the value of the cause.
- B. changes the relationship between the cause and its effects.
- C. always changes the value of the cause.
- D. None of the above.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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D - Reflect and discuss – Intervening on Causes 1

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- A. determines the value of the cause.
- B. changes the relationship between the cause and its effects.
- C. always changes the value of the cause.
- D. None of the above.

Answer - A.

Not B: An ideal intervention on a cause does not, in general, change the relationship between the cause and its effect.

Not C: Ideal interventions need not CHANGE the value of a cause - they ensure that the cause will take on the value intended.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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E - Reflect and discuss – Intervening on Effects 1

 **Question No 08** – Let's assume that high cholesterol causes poor artery condition. **If we ideally intervene on the condition of your arteries, then... (choose all that apply)**

- A. the relationship between Cholesterol Level and Artery Condition changes, at least temporarily.
- B. the value of Cholesterol Level changes.
- C. the value of Artery Condition is always changed.

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- A. the relationship between Cholesterol Level and Artery Condition changes, at least temporarily.**
- B. the value of Cholesterol Level changes.
- C. the value of Artery Condition is always changed.

Answer - A.

Not B: Since Cholesterol Level is a cause of Artery Condition, and we have intervened on Artery Condition, we have no effect on the value of Cholesterol Level.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

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E - Reflect and discuss – Intervening on Effects 1

 **Question No 09** – Assuming that there is no cycle between causes and effects, then in general, **an ideal intervention on an effect...**
(choose all that apply)

- A. does nothing to the value of the causes of the effect.
- B. changes the relationship between the effect and its causes.
- C. None of the above.

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- B. changes the relationship between the effect and its causes.
- C. None of the above.

Answer – A and B.

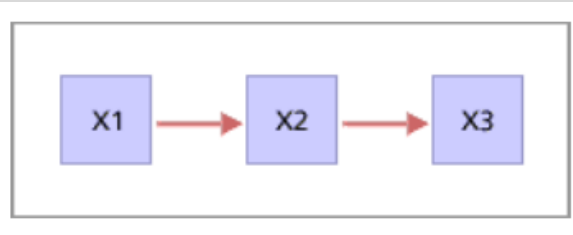
An ideal intervention does nothing to the value of the causes of an effect, but it changes the relationship between an effect and its causes.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

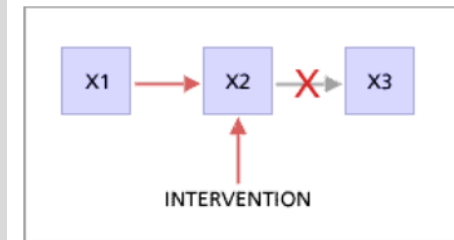
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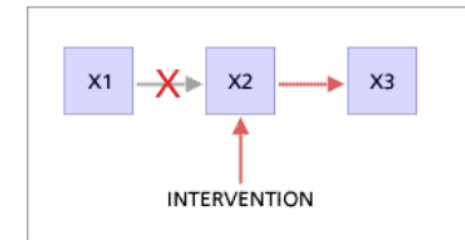
Question No 10 – Suppose that the following causal graph represents the pre-manipulated causal system governing the variables X_1 , X_2 , and X_3 .



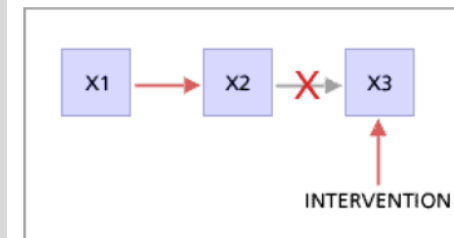
Which of the following causal graphs represents the post-manipulation system following an ideal intervention on X_2 ?



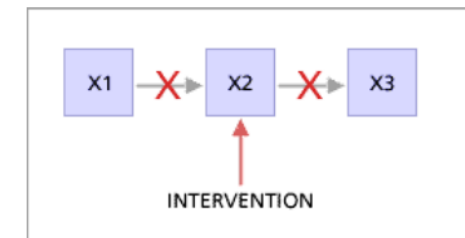
A.



B.



C.



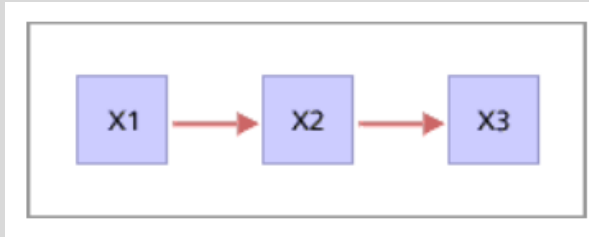
D.

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E - Reflect and discuss – Intervening on Effects 1

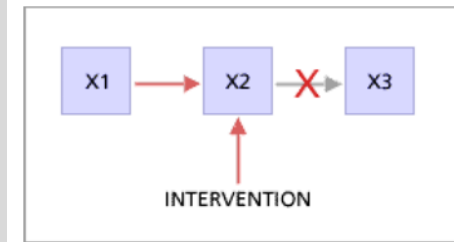
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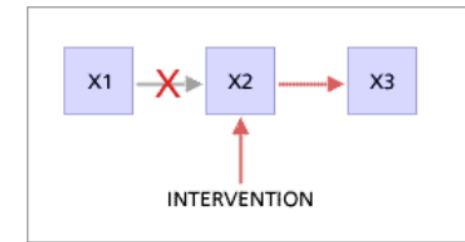
Which of the following causal graphs represents the post-manipulation system following an ideal intervention on X_2 ?

Answer - B.

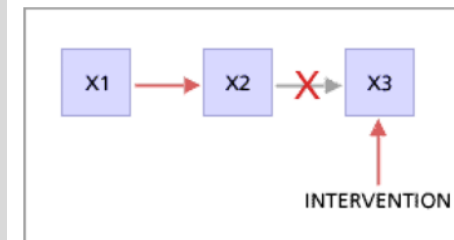
- Not A: Interventions on an effect change the relationship between the effect and its causes. The intervention on X_2 changes its relationship with its causes, which in this case are X_1 . Graph A mistakenly leaves the relationship between X_1 and X_2 intact.
- Not C: Graph C represents an intervention on X_3 , not X_2 .
- Not D: Interventions on an effect change the relationship between the effect and its causes. The intervention on X_2 changes its relationship with its causes, which in this case are X_1 . In graph D, the relationship between X_2 and X_3 has been changed, but X_3 is an effect of X_2 , not a cause.



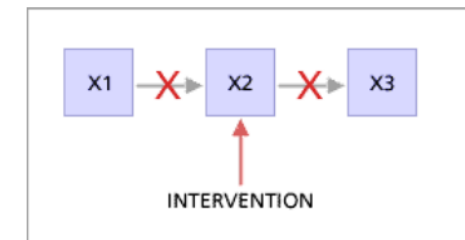
A.



B.



C.



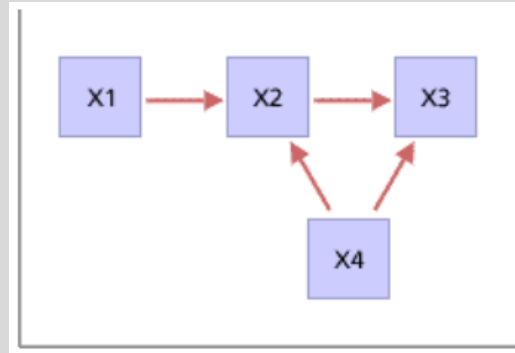
D.

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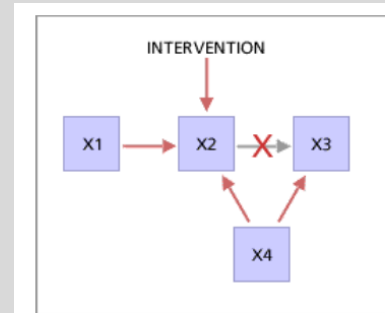
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E - Reflect and discuss – Intervening on Effects 1

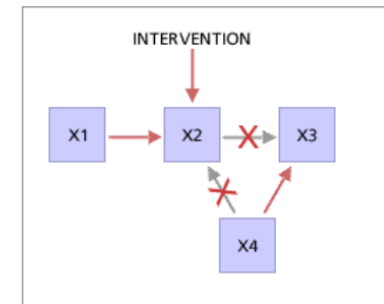
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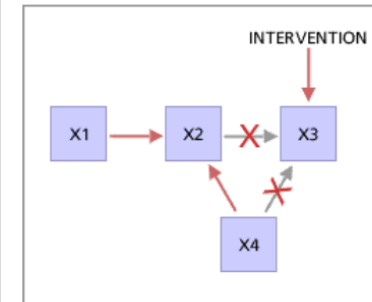
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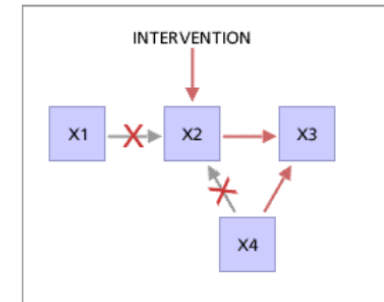
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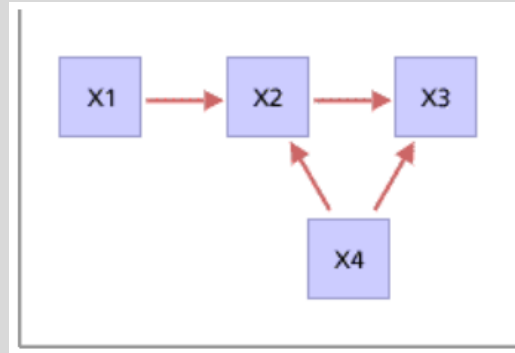
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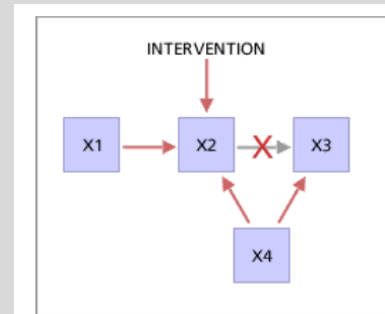
Question No 11 – Suppose that the following causal graph represents the pre-manipulated causal system governing the variables X_1 , X_2 , X_3 , and X_4 .



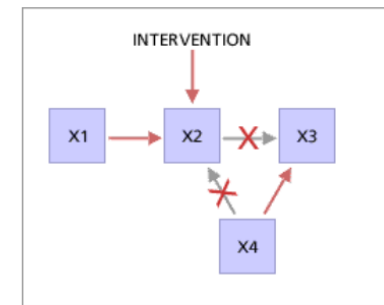
Which of the following causal graphs represents the post-manipulation system following an ideal intervention on X_2 ?

Answer - D.

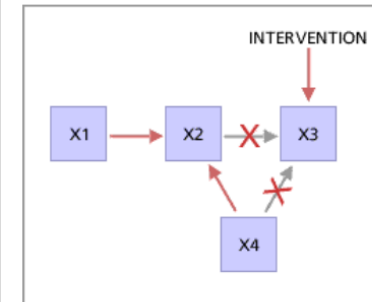
- Not A: The post-manipulated graph in A correctly breaks the edge from X_4 , but it does not break the edge from X_1 , and incorrectly breaks the edge from X_2 to X_3 .
- Not B: Graph B includes a change to the relationship between X_2 and one of its effects, which is incorrect.
- Not C: Graph C represents an intervention on X_3 , not X_2 .



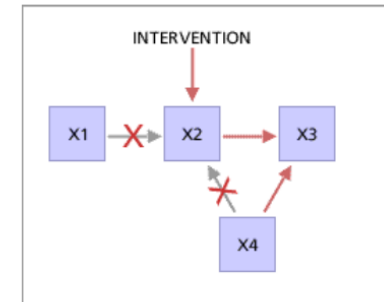
A.



B.



C.



D.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

 **Question No 12** – Consider the following causal system representing a bicycle: {Gear selected, Pedals moving}. **Is the variable ‘Gear selected’ dependent, or independent? (choose one)**

- A. Independent
- B. Dependent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

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A. Independent

B. Dependent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

 **Question No 13** – Consider the following causal system describing the causal relations inside an automobile: {Gear shift setting, Accelerator pedal setting, Engine RPMs}. **Is the variable Engine RPMs (revolutions per minute of the engine) dependent or independent? (choose one)**

- A. Dependent
- B. Independent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

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B. Independent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

 **Question No 14** – Consider the solar system, with variables ‘distance of the Earth from the sun’, ‘temperature of the sun’, and ‘temperature of the Earth’. **Is the variable ‘temperature of the Earth’ dependent, or independent? (choose one)**

- A. Dependent
- B. Independent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

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A. Dependent

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Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

 **Question No 15** – Consider a following system: a human body, with variables ‘staph infection’, ‘fever’, and ‘body temperature’. **Is the variable ‘body temperature’ dependent or independent? (choose one)**

- A. Dependent
- B. Independent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

Question No 15 – Consider a following system: a human body, with variables ‘staph infection’, ‘fever’, and ‘body temperature’. **Is the variable ‘body temperature’ dependent or independent? (choose one)**

A. Dependent

B. Independent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

Question No 16 – Consider the electrical system of a house, which includes variables {Lamp [on, off], Refrigerator [on, off], Radio [on, off], Lamp Switch [up, down]}. **Is the variable ‘Lamp Switch’ dependent, or independent? (choose one)**

- A. Independent
- B. Dependent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

G - Reflect and discuss – Exogenous and Endogenous Variables 1

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A. Independent

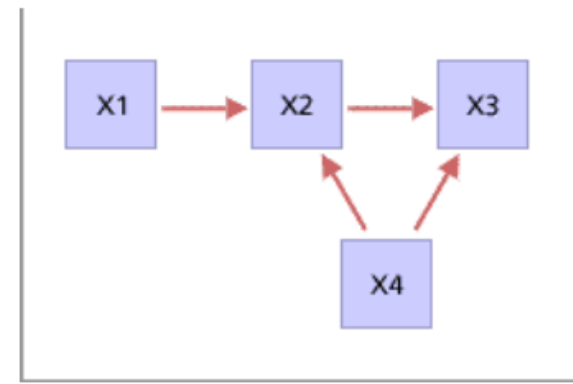
B. Dependent

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Exogenous and Endogenous Variables 2

Question No 17 – Use this graph to answer the following question:



For which of the following variables would an intervention on it be structure-altering? (choose all that apply)

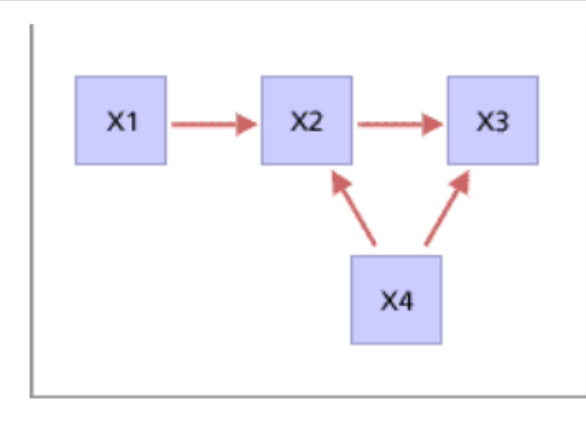
- A. X1.
- B. X2.
- C. X3.
- D. X4.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

H - Reflect and discuss – Exogenous and Endogenous Variables 2

Question No 17 – Use this graph to answer the following question:



For which of the following variables would an intervention on it be structure-altering? (choose all that apply)

- A. X1.
- B. X2.
- C. X3.
- D. X4.

Answer - B and C

X2 and X3 are endogenous variables, and thus intervening to change either of them is a structure-altering intervention.

Not A: X1 is an exogenous variable, and thus intervening to change it is a structure-preserving intervention.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Interventions on a Cyclic System 1

 **Question No 18** – Consider the causal system of the bicycle in the simulation to have three variables:

1. Pedals [moving, still]
2. Rear Wheel [moving, still]
3. Gear Shift [forward, back]

Choose all of the following that are direct causes in this system.

- A. Pedal $\rightarrow$ Rear Wheel.
- B. Rear Wheel $\rightarrow$ Pedal.
- C. Pedal $\rightarrow$ Gear Shift.
- D. Gear Shift $\rightarrow$ Pedal.
- E. Rear Wheel $\rightarrow$ Gear shift.
- F. Gear Shift $\rightarrow$ Rear Wheel.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

I - Reflect and discuss – Interventions on a Cyclic System 1

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1. Pedals [moving, still]
2. Rear Wheel [moving, still]
3. Gear Shift [forward, back]

Choose all of the following that are direct causes in this system.

- A. Pedal → Rear Wheel.**
- B. Rear Wheel → Pedal.
- C. Pedal → Gear Shift.
- D. Gear Shift → Pedal.
- E. Rear Wheel → Gear shift.
- F. Gear Shift → Rear Wheel.**

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Interventions on a Cyclic System 2

 **Question No 19** – Consider the causal system of the bicycle in the second simulation to have three variables:

1. Pedals [moving, still]
2. Rear Wheel [moving, still]
3. Gear Shift [forward, back]

Choose all of the following that are direct causes in this system.

- A. Pedal $\rightarrow$ Rear Wheel.
- B. Rear Wheel $\rightarrow$ Pedal.
- C. Pedal $\rightarrow$ Gear Shift.
- D. Gear Shift $\rightarrow$ Pedal.
- E. Rear Wheel $\rightarrow$ Gear shift.
- F. Gear Shift $\rightarrow$ Rear Wheel.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

J - Reflect and discuss – Interventions on a Cyclic System 2

Question No 19 – Consider the causal system of the bicycle in the second simulation to have three variables:

1. Pedals [moving, still]
2. Rear Wheel [moving, still]
3. Gear Shift [forward, back]

Choose all of the following that are direct causes in this system.

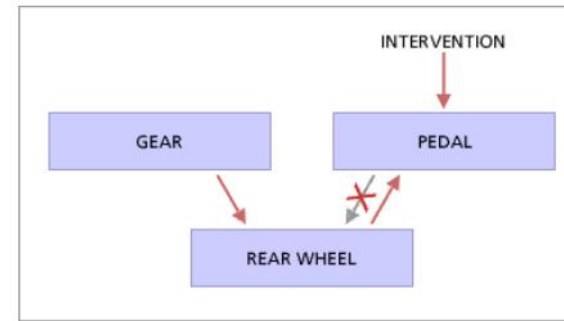
- A. Pedal → Rear Wheel.**
- B. Rear Wheel → Pedal.**
- C. Pedal → Gear Shift.
- D. Gear Shift → Pedal.
- E. Rear Wheel → Gear shift.
- F. Gear Shift → Rear Wheel.**

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

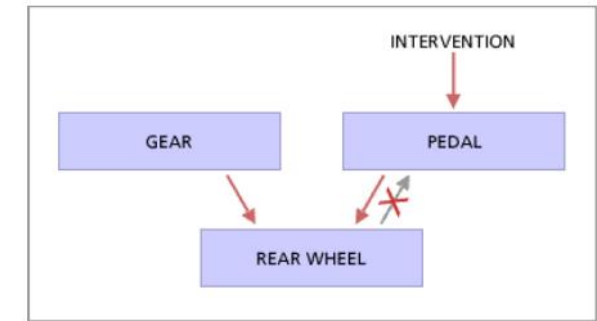
Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Interventions on a Cyclic System 3

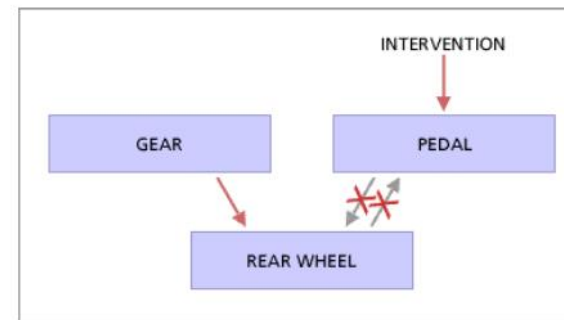
Question No 20 – Which of the following is the post-manipulation graph for an intervention on the pedals? (choose one)



A.



B.



C.

None of these.

D.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

K - Reflect and discuss – Interventions on a Cyclic System 3

Question No 20 – Which of the following is the post-manipulation graph for an intervention on the pedals? (choose one)

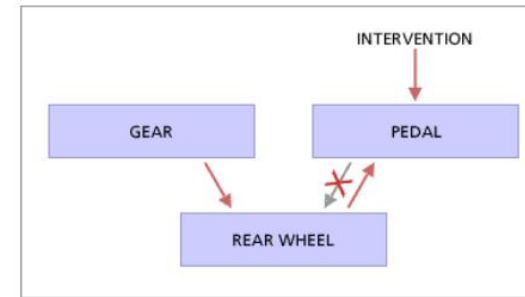
Answer - B

Interventions change the relationship between an effect and its causes, but leave intact the relationship between a cause and its effect.

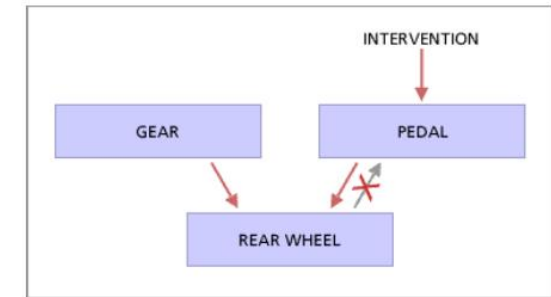
Not A: In the edge broken in the graph you chose, Pedal is a cause. But interventions on causes don't break edges from the cause to any of its effects.

Not C: Both the edge from Pedal to Rear Wheel and the edge from Rear Wheel to Pedal are broken in this graph, but interventions on causes don't break edges from the cause to its effects. So we should leave the edge from Pedal to Rear Wheel intact.

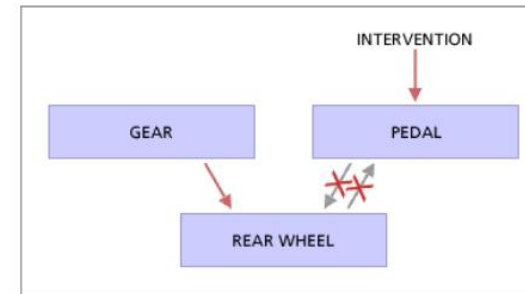
Not D: One of the choices is correct. Recall that interventions change the relationship between an effect and its causes, but leave intact the relationship between a cause and its effect.



A.



B.



C.

None of these.

D.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Case Study 1

 **Question No 21 – Which of the following variables was most directly under the control of the city officials of Cali and Bogotá? (choose one)**

- A. Homicide Rate
- B. Number of People Carrying Guns
- C. Crime Rate
- D. Frequency of Gun Searches

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Case Study 1

 **Question No 21 – Which of the following variables was most directly under the control of the city officials of Cali and Bogotá? (choose one)**

- A. Homicide Rate
- B. Number of People Carrying Guns
- C. Crime Rate
- D. Frequency of Gun Searches**

Answer - D

- Not A: Although the Homicide Rate was the ultimate target of the intervention, the officials could only intervene directly on the frequency of gun searches, which they hoped would reduce the number of people carrying guns, which in turn would reduce the Homicide Rate.
- Not B: Although the officials hoped that by increasing the frequency of gun searches, they would reduce the number of people carrying guns, they could most directly change only the frequency of searches.
- Not C: Although the Crime Rate, more particularly the Homicide Rate, was the ultimate target of the intervention, the officials could only intervene directly on the frequency of gun searches, which they hoped would reduce the number of people carrying guns, which in turn would reduce the Homicide Rate.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Case Study 1

 **Question No 22** – Relative to the following variables:

- Frequency of Searches
- Homicide Rate
- Number of Police Patrolling
- Number of People Carrying Guns

The gun ban would be a fat hand intervention if:

- A. It increased the number of homicides
- B. It directly increased both the frequency of gun searches and the number of police patrolling the streets
- C. It reduced the number of people carrying guns but not the number of homicides
- D. It reduced the number of people carrying guns, which reduced number of homicides.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

L - Reflect and discuss – Case Study 1

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- C. It reduced the number of people carrying guns but not the number of homicides
- D. It reduced the number of people carrying guns, which reduced number of homicides.

Answer - B

- Not A: This choice describes the chain of causation that resulted from the policy, but doesn't bear on whether the particular version of a gun ban implemented in Columbia in this article was a fat hand intervention or not.
- Not C: This choice describes the ultimate result of the policy, but doesn't bear on whether the particular version of a gun ban implemented in Columbia in this article was a fat hand intervention or not.
- Not D: This choice describes the hoped for chain of causation, but doesn't bear on whether the particular version of a gun ban implemented in Columbia in this article was a fat hand intervention or not.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Case Study 2

 **Question No 23 – Which of the following variables was most directly under the control of the researchers in the study described in the article? (choose one)**

- A. Participation in TV and Video-Game Reduction Lessons
- B. Number of Hours Spent Playing Video Games
- C. Diet
- D. Number of Hours Spent Watching TV

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Case Study 2

 Question No 23 – Which of the following variables was most directly under the control of the researchers in the study described in the article? (choose one)

- A. Participation in TV and Video-Game Reduction Lessons
- B. Number of Hours Spent Playing Video Games
- C. Diet
- D. Number of Hours Spent Watching TV

Answer - A

- Not B: Although the researchers want to affect the number of hours spent playing video games, they could only directly affect whether or not subjects participated in TV and Video-Game reduction lessons.
- Not C: Although the researchers ultimately want to improve the diet of children, they only directly affect whether or not children participated in TV and Video-Game reduction lessons, which they hoped would reduce hours spent playing watching TV and playing video games, which in turn would affect diet.
- Not D: Although the researchers want to affect the number of hours spent watching TV, they could only directly affect whether or not subjects participated in TV and Video-Game reduction lessons.

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Case Study 2

 **Question No 24** – Consider the following variables:

1. Time spent watching TV
2. Sensitivity towards other children
3. School Performance

Participation in the TV reduction lessons would be a fat hand intervention if:

- A. It directly affects both time spent watching TV and sensitivity towards other children
- B. It directly decreases the amount of time watching TV and thereby indirectly decreases aggressive behavior
- C. It indirectly improves both school performance and the child-parent relationship
- D. It increases aggressive behavior

Reading Exercise – No 17 (Module 5 | Causation - Interventions)

Instructions/information: You may refer to your notes as you answer them. You are encouraged to reflect & discuss with your group mates

M - Reflect and discuss – Case Study 2

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- B. It directly decreases the amount of time watching TV and thereby indirectly decreases aggressive behavior
- C. It indirectly improves both school performance and the child-parent relationship
- D. It increases aggressive behavior

Answer - A

- Not B: This choice only informs us of its direct effect on the variable it targeted as well as the indirect effect.
- Not C: Knowing the interventions indirect effects don't tell you whether its direct effect targeted more than one variable.